

MA_vax: vaccination-impact analysis report

Massachusetts, 2025–2026 influenza season. Age-structured SEIR model with a parallel vaccinated arm, calibrated to daily hospital admissions by age group via Bayesian MCMC. This report documents the model, the fit, and the resulting vaccination-impact analysis.

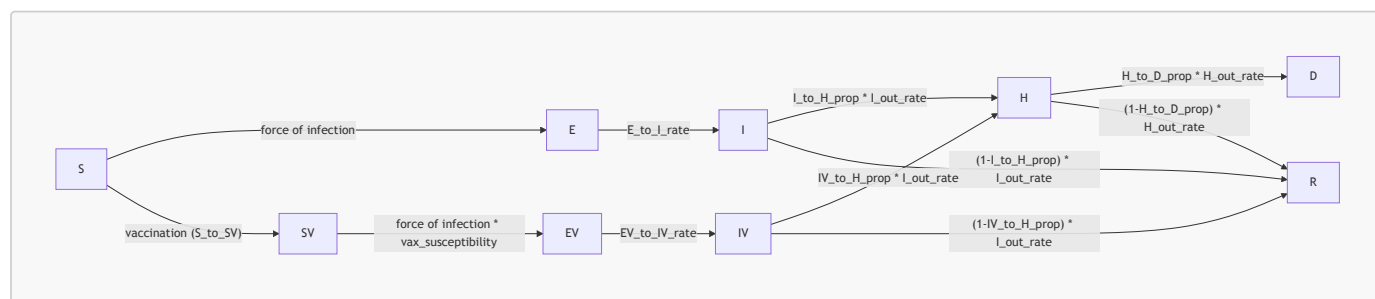
1. Model structure

1.1 Compartments

Nine compartments, each stratified by 7 age groups (0, 1–4, 5–12, 13–17, 18–49, 50–64, 65+):

Compartment	Meaning
S	Susceptible, unvaccinated
E	Exposed (latent), unvaccinated
I	Infectious, unvaccinated
R	Recovered/removed (either arm)
SV	Susceptible, vaccinated
EV	Exposed (latent), vaccinated
IV	Infectious, vaccinated
H	Hospitalized (either arm)
D	Dead (either arm)

The vaccinated arm (SV → EV → IV → H/R) mirrors the unvaccinated arm (S → E → I → H/R), with its own susceptibility and severity parameters. H, R, D are shared, pooled compartments — once hospitalized (or recovered), an individual's vaccination history is no longer tracked.



1.2 Force of infection

Contacts come from fixed age×age contact matrices (Mistry et al. 2021 synthetic contact matrices), with school/work contacts removed on non-school/work days:

```

C(t) = total_C - (1 - is_school(t))·school_C - (1 - is_work(t))·work_C
beta_adj(t) = beta_baseline · m(t) · (1 + humidity_impact · exp(-180 · humidity(t)))
wtd_inf_prop(t) = (I·I_relative_infectiousness + IV·IV_relative_infectiousness) / population
foi(t) = beta_adj(t) · (C(t) @ wtd_inf_prop(t))
S_to_E = foi(t) · relative_suscept · S
SV_to_EV = foi(t) · vax_susceptibility · SV
  
```

$m(t)$ is a smoothly time-varying transmission multiplier (§2.3) that absorbs behavioral/seasonal variation the mechanistic terms above don't otherwise capture. $IV_relative_infectiousness = 1.0$ — breakthrough infections (in vaccinated individuals)

are assumed exactly as transmissible as unvaccinated infections. `vax_susceptibility` (age-specific, ≤ 1) is the residual susceptibility of a vaccinated individual; $1 - \text{vax_susceptibility}$ is the model's implied vaccine effectiveness (VE) against infection.

1.3 Progression, hospitalization, death

$$\begin{aligned} E_to_I &= E_to_I_rate \cdot E & EV_to_IV &= EV_to_IV_rate \cdot EV \\ I_to_H &= I_out_rate \cdot I_to_H_prop \cdot I & IV_to_H &= I_out_rate \cdot IV_to_H_prop \cdot IV \\ I_to_R &= I_out_rate \cdot (1-I_to_H_prop) \cdot I & IV_to_R &= I_out_rate \cdot (1-IV_to_H_prop) \cdot IV \\ H_to_D &= H_out_rate \cdot H_to_D_prop \cdot H \\ H_to_R &= H_out_rate \cdot (1-H_to_D_prop) \cdot H \end{aligned}$$

$IV_to_H_prop < I_to_H_prop$ for every age group — vaccination reduces hospitalization risk *given* infection (severity VE), on top of reducing infection risk.

1.4 Vaccination flow

Daily doses (an age-specific proportion of the population, with a 14-day delay between dose and effective immunity) are applied as an exact $S \rightarrow SV$ count each day, capped at whatever remains in S :

$$\begin{aligned} \text{base}(t) &= S + SV \\ S_to_SV(t) &= \min(\text{round}(\text{vax_prop}(t) \cdot \text{base}(t)), S) \end{aligned}$$

The base pool ($S + SV$) is not eroded by vaccination itself — only infection depletes it — so a roughly flat input proportion vaccinates a roughly constant head-count per day, until S starts running low late in the season.

1.5 Numerical scheme

Deterministic simulations use explicit Euler integration with 7 sub-steps per day. Stochastic simulations (used for confidence intervals throughout this report) use a chain-binomial scheme at the same sub-daily resolution. Vaccination itself stays a deterministic scheduled count in both, since it comes from an external delivery schedule rather than a hazard rate.

1.6 Initial conditions

At the start of the simulation (2025-09-01): $S = \text{population} - E_0$, $E = E_0$ (age-specific seed counts, scaled by a single fitted multiplier, §2.3), and all other compartments start at zero.

2. Parameters

2.1 Fixed scalar parameters

Parameter	Value	Meaning
<code>num_days</code>	250	Simulation length (through ~2026-05-08)
<code>relative_suscept</code>	1.0	Susceptibility multiplier, unvaccinated arm
<code>I_relative_infectiousness</code>	1.0	Infectiousness weight, unvaccinated I
<code>IV_relative_infectiousness</code>	1.0	Infectiousness weight, vaccinated IV
<code>E_to_I_rate</code>	0.5 /day	~2-day latent period
<code>EV_to_IV_rate</code>	0.5 /day	Same, vaccinated arm
<code>I_out_rate</code>	0.333 /day	~3-day infectious period
<code>H_out_rate</code>	0.17 /day	~6-day hospital stay
<code>vax_transfer_delay_days</code>	14	Days from dose to modeled immunity

2.2 Age-stratified fixed parameters

Age group	Population	Hospitalization risk, given infection ¹	Hospitalization risk, given breakthrough infection ¹	Death risk, given hospitalization	Residual susceptibility if vaccinated	Initial infections seeded ¹
0	70,067	0.697%	0.636%	1.74%	0.57	2
1-4	280,268	0.697%	0.636%	1.74%	0.57	8
5-12	606,291	0.274%	0.250%	1.17%	0.57	17
13-17	411,782	0.274%	0.250%	1.17%	0.57	12
18-49	2,978,204	0.561%	0.511%	2.63%	0.79	85
50-64	1,424,434	1.060%	0.966%	6.30%	0.79	41
65+	1,221,349	9.091%	6.273%	7.99%	1.00	35

¹ These are pre-fit baseline values — the calibration scales both hospitalization-risk columns by a fitted age-specific multiplier (§2.3), so the values actually used in the calibrated model are these figures × that multiplier and the vaccine effectiveness against hospitalization remains the same. A residual susceptibility of 1.00 for 65+ means the fitted baseline assumes **no infection-blocking effect** of vaccination in that age group (only the severity effect applies there). This holds for the **Low VE** sensitivity scenario too, but not for **High VE**, which scales it to 0.86 — see Table S.A.4.

2.3 Fitting method

Free parameters were estimated with an affine-invariant ensemble Markov Chain Monte Carlo sampler (40 walkers, 4000 iterations per walker), run in parallel. The free parameters and their priors:

Parameter	Prior	Role
beta_baseline	Uniform(0.015, 0.06)	Baseline transmission rate
humidity_impact	Uniform(0, 1)	Strength of humidity forcing
Initial-seed multiplier	Log-uniform(0.1, 10)	Multiplies the age-specific initial-infection seed counts
Hospitalization-risk multiplier (× 7, one per age group)	Uniform(0.1, 2.0)	Multiplies both hospitalization-risk columns (§2.2) for that age group
m(t) log-increments (× 18, one per 14-day knot)	Normal(0, 0.25)	Month-to-month-ish random-walk steps in log-transmission (§1.2)
phi	—	Negative-Binomial dispersion (likelihood nuisance parameter, not a model input)

Likelihood: a Negative-Binomial (NB2) observation model, chosen over Poisson to allow overdispersion, jointly across 8 targets — daily hospital admissions **by age group** (7 time series) plus a single scalar **end-of-season cumulative hospitalizations by age** target.

Posterior sampling: the first 1700 iterations were discarded as burn-in, and the remaining chain thinned to every 200th sample, leaving 638 posterior draws. Two point estimates are used in this report:

- **Posterior mean:** the marginal mean of each parameter. Cheap and stable, but for a correlated posterior can land on a parameter *combination* the sampler never actually visited.
- **"Best" point:** the single posterior draw with the highest log-posterior — not the mean of anything, so it preserves whatever correlation structure exists between parameters.

These two can disagree materially (§2.4) — a reminder that the marginal mean of a correlated posterior needn't correspond to a jointly plausible parameter combination. **m(t)**'s random-walk prior is a smoothness assumption, not a mechanistic one — it

should not be over-interpreted as an independently-measured behavioral signal, only as "how much transmission needs to have moved, beyond what humidity/contacts explain, to reconcile the model with the data."

2.4 Fitted parameters

Posterior mean $\pm$ 90% credible interval (5th–95th percentile) across the 638 posterior draws, vs. the "best" (highest-posterior-density) point:

Parameter	Posterior mean	5%	95%	"Best" point
beta_baseline	0.0367	0.0314	0.0419	0.0379
humidity_impact	0.595	0.297	0.910	0.876
Initial-seed multiplier	1.80	0.58	3.98	1.37
Hospitalization-risk multiplier — 0	1.45	0.94	1.91	1.67
Hospitalization-risk multiplier — 1-4	1.63	1.16	1.96	1.76
Hospitalization-risk multiplier — 5-12	1.31	0.86	1.82	1.22
Hospitalization-risk multiplier — 13-17	0.96	0.62	1.41	0.97
Hospitalization-risk multiplier — 18-49	0.52	0.34	0.70	0.47
Hospitalization-risk multiplier — 50-64	0.70	0.46	0.96	0.65
Hospitalization-risk multiplier — 65+	0.93	0.66	1.24	0.94
phi (NB dispersion)	116	25	357	254

The 18 **m(t)** log-increments aren't independently interpretable as a table — they're summarized visually through the fit check in §4 instead.

2.5 Cumulative vaccination coverage

Cumulative proportion of each age group actually vaccinated over the season (reported vaccination-schedule data, summed over the season):

Age group	Population	Cumulative coverage
0	70,067	45.4%
1-4	280,268	90.7%
5-12	606,291	71.4%
13-17	411,782	55.3%
18-49	2,978,204	40.5%
50-64	1,424,434	60.4%
65+	1,221,349	73.2%
All (population-weighted)	6,992,395	55.9%

Coverage spans a wide range: four of the seven age groups sit below a 70% mark — 18-49 lowest at 40.5%, then 0 (45.4%), 13-17 (55.3%) and 50-64 (60.4%) — while **1-4 (90.7%) is well above it, and 5-12 (71.4%) and 65+ (73.2%) are just above it.** This matters for how the "scale to 70% coverage" rows in the appendix (Table S.A.3/S.A.6) should be read: only the four groups below the mark are scaled up, and the three already above it are left exactly as they are, contributing all-zero columns.

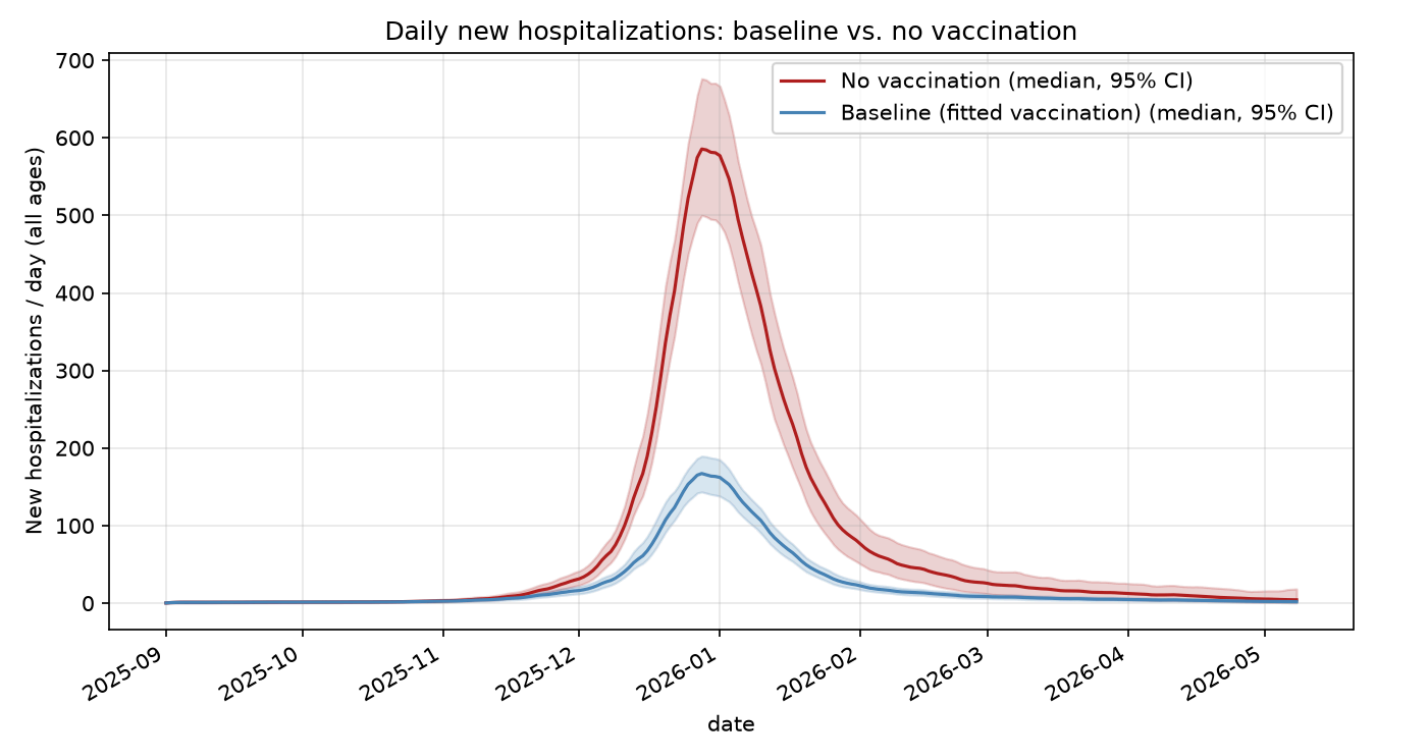
The figures above are what the vaccination **schedule** reports. The doses that actually land are fewer — the §1.4 cap declines to vaccinate someone who has already been infected — putting simulated coverage 0.6 to 2.3 percentage points lower per age group. Those undelivered doses are not saved: in the real world they are still bought and administered, into arms that no longer benefit, at least in our model where the recovered and deceased compartments are final. The appendix reconciles the two figures age group by age group, and every "per 100,000 doses" panel in this report divides by the **scheduled** count for that reason.

3. Vaccination-impact results

Every table in this section reports a **median and 95% interval across 638 simulations** — one simulation per posterior parameter draw (§2.3), re-using the same draw for every scenario being compared within a table so that the comparison is paired (variance from the parameter draw itself cancels out of the *difference* between scenarios). These intervals therefore reflect **calibration uncertainty** — how much the vaccination- impact conclusions would change under a different (but similarly plausible) fit to the same data — not day-to-day epidemic-process noise.

New daily hospitalizations: baseline vs. no vaccination

Total-population new hospitalizations per day, posterior median and 95% interval across the 638 parameter draws (§2.3, same posterior-uncertainty basis as the rest of this section), comparing the fitted baseline vaccination schedule against a counterfactual with no vaccination at all:



The fitted vaccination program cuts the peak in daily new hospitalizations to roughly **a third** of what it would otherwise be (median peak ≈ 165 /day baseline vs. ≈ 585 /day with no vaccination, both around 2025-12-28) and reduces the epidemic's height without changing its timing.

Table S.A.1 — Hospitalizations averted, infection vs. severity protection

Decomposes the total hospitalizations averted by vaccination into two channels: protection against getting infected at all, and — *given* a breakthrough infection still happens — protection against that infection becoming severe enough to need hospitalization.

Infection protection (no vaccination $\rightarrow$ infection-protection-only, i.e. VE against infection retained but VE against severity zeroed out)

Age group	% Hospitalizations Averted	Averted per 100,000 Population	Averted per 100,000 Doses
0	69.0% [63.8% – 72.5%]	116.8 [75.1 – 152.6]	257.6 [165.6 – 336.5]
1-4	74.5% [70.0% – 77.6%]	168.4 [125.3 – 203.0]	185.7 [138.1 – 223.7]
5-12	71.8% [66.2% – 75.6%]	70.3 [47.2 – 94.5]	98.4 [66.1 – 132.2]
13-17	69.1% [63.0% – 73.2%]	49.9 [33.4 – 73.3]	90.2 [60.4 – 132.5]
18-49	64.0% [57.6% – 68.3%]	44.7 [31.9 – 59.3]	110.4 [78.7 – 146.2]

Age group	% Hospitalizations Averted	Averted per 100,000 Population	Averted per 100,000 Doses
50-64	65.4% [59.5% – 69.4%]	102.8 [72.9 – 136.3]	170.3 [120.7 – 225.9]
65+	63.0% [57.1% – 67.0%]	701.6 [542.4 – 852.1]	958.4 [740.9 – 1163.9]
All	64.1% [58.4% – 68.1%]	180.3 [141.2 – 214.5]	322.7 [252.6 – 383.8]

Severity protection (infection-protection-only → full baseline, i.e. adding back VE against severity)

Age group	% Hospitalizations Averted	Averted per 100,000 Population	Averted per 100,000 Doses
0	0.6% [0.6% – 0.8%]	1.1 [0.7 – 1.6]	2.4 [1.6 – 3.4]
1-4	1.3% [1.1% – 1.5%]	2.9 [2.0 – 3.9]	3.2 [2.2 – 4.3]
5-12	1.0% [0.9% – 1.2%]	1.0 [0.7 – 1.3]	1.4 [1.0 – 1.8]
13-17	0.9% [0.7% – 1.0%]	0.6 [0.4 – 0.9]	1.1 [0.8 – 1.6]
18-49	0.8% [0.7% – 0.9%]	0.6 [0.5 – 0.7]	1.4 [1.1 – 1.7]
50-64	1.2% [1.1% – 1.4%]	1.9 [1.5 – 2.3]	3.2 [2.5 – 3.9]
65+	6.7% [6.0% – 7.7%]	74.4 [67.5 – 81.1]	101.6 [92.2 – 110.8]
All	4.9% [4.4% – 5.8%]	13.9 [12.7 – 15.0]	24.9 [22.7 – 26.9]

Total (no vaccination → full baseline)

Age group	% Hospitalizations Averted	Averted per 100,000 Population	Averted per 100,000 Doses
0	69.6% [64.6% – 73.0%]	117.9 [76.0 – 154.0]	259.9 [167.7 – 339.5]
1-4	75.8% [71.5% – 78.8%]	171.2 [127.5 – 206.9]	188.8 [140.6 – 228.1]
5-12	72.8% [67.4% – 76.5%]	71.3 [48.1 – 95.8]	99.8 [67.4 – 134.1]
13-17	70.0% [64.1% – 74.0%]	50.5 [33.9 – 74.1]	91.3 [61.3 – 134.0]
18-49	64.8% [58.6% – 69.0%]	45.3 [32.4 – 59.9]	111.7 [80.0 – 147.7]
50-64	66.6% [60.9% – 70.5%]	104.7 [74.6 – 138.6]	173.5 [123.6 – 229.7]
65+	69.7% [64.8% – 72.9%]	776.3 [621.4 – 927.4]	1060.4 [848.8 – 1266.8]
All	69.1% [64.1% – 72.5%]	194.3 [154.7 – 228.8]	347.6 [276.8 – 409.4]

Almost all of the averted burden comes from **blocking infection**, not from reducing severity given a breakthrough.

Table S.A.2 — Hospitalizations averted by age group vaccinated

Each column vaccinates a single age group only (all others left unvaccinated) and compares to no vaccination at all; "All" is the full baseline schedule. Rows are the age group in which hospitalizations are counted, so off-diagonal cells show the indirect (transmission-blocking) benefit to *other* age groups from vaccinating this one.

% reduction in hospitalizations

Age group (counted)	0 vaccinated	1-4 vaccinated	5-12 vaccinated	13-17 vaccinated	18-49 vaccinated	50-64 vaccinated	65+ vaccinated	All vaccinated
0	16.9% [16.3% – 17.3%]	8.5% [7.4% – 9.5%]	24.4% [20.7% – 27.3%]	14.1% [11.8% – 16.0%]	18.1% [15.3% – 20.3%]	8.8% [7.4% – 10.0%]	0.0% [-0.0% – 0.0%]	69.6% [64.6% – 73.0%]
1-4	0.5% [0.4% – 0.5%]	38.8% [37.0% – 40.6%]	25.8% [21.8% – 29.8%]	13.4% [11.1% – 15.7%]	16.8% [14.0% – 19.6%]	8.3% [6.8% – 9.8%]	0.0% [-0.0% – 0.0%]	75.8% [71.5% – 80.1%]

Age group (counted)	0 vaccinated	1-4 vaccinated	5-12 vaccinated	13-17 vaccinated	18-49 vaccinated	50-64 vaccinated	65+ vaccinated	All vaccinated
		40.2%]	29.0%]	15.4%]	19.1%]	9.5%]	0.0%]	78.8%]
5-12	0.4% [0.3% – 0.4%]	7.1% [5.8% – 8.1%]	46.6% [42.5% – 49.7%]	13.1% [10.5% – 15.2%]	14.3% [11.4% – 16.8%]	7.3% [5.8% – 8.6%]	0.0% [-0.0% – 0.0%]	72.8% [67.4% – 76.5%]
13-17	0.3% [0.3% – 0.4%]	5.8% [4.6% – 6.8%]	21.5% [17.3% – 24.9%]	35.1% [32.0% – 37.5%]	14.4% [11.4% – 16.9%]	7.7% [6.1% – 9.0%]	0.0% [-0.0% – 0.0%]	70.0% [64.1% – 74.0%]
18-49	0.4% [0.4% – 0.5%]	7.0% [5.8% – 8.0%]	22.2% [18.4% – 25.4%]	13.6% [11.2% – 15.7%]	23.7% [20.7% – 26.0%]	9.1% [7.5% – 10.4%]	0.0% [-0.0% – 0.0%]	64.8% [58.6% – 69.0%]
50-64	0.4% [0.3% – 0.4%]	6.6% [5.5% – 7.6%]	22.0% [18.3% – 25.0%]	14.0% [11.6% – 16.0%]	17.1% [14.3% – 19.4%]	21.3% [19.5% – 22.7%]	0.0% [-0.0% – 0.0%]	66.6% [60.9% – 70.5%]
65+	0.4% [0.4% – 0.5%]	7.3% [6.2% – 8.1%]	23.5% [20.0% – 26.3%]	14.6% [12.4% – 16.5%]	17.6% [14.9% – 19.7%]	10.4% [8.9% – 11.6%]	18.3% [18.1% – 18.5%]	69.7% [64.8% – 72.9%]
All	0.5% [0.5% – 0.6%]	8.2% [7.2% – 8.9%]	23.9% [20.2% – 26.8%]	14.7% [12.3% – 16.6%]	18.0% [15.2% – 20.2%]	11.3% [9.6% – 12.6%]	12.7% [12.1% – 13.3%]	69.1% [64.1% – 72.5%]

Hospitalizations averted per 100,000 population

Age group (counted)	0 vaccinated	1-4 vaccinated	5-12 vaccinated	13-17 vaccinated	18-49 vaccinated	50-64 vaccinated	65+ vaccinated	All vaccinated
0	28.6 [18.5 – 37.4]	14.5 [8.9 – 18.9]	41.4 [25.4 – 53.9]	23.9 [14.4 – 31.2]	30.7 [18.8 – 40.1]	14.9 [9.1 – 19.6]	0.0 [-0.0 – 0.0]	117.9 [76.0 – 154.0]
1-4	1.1 [0.8 – 1.3]	87.6 [65.2 – 105.9]	58.6 [41.7 – 69.4]	30.4 [21.5 – 36.0]	38.0 [27.0 – 45.1]	18.8 [13.1 – 22.3]	0.0 [-0.0 – 0.0]	171.2 [127.5 – 206.9]
5-12	0.4 [0.2 – 0.5]	6.9 [4.3 – 9.6]	45.5 [30.3 – 61.4]	12.7 [7.8 – 18.0]	14.0 [8.5 – 19.9]	7.1 [4.3 – 10.1]	0.0 [-0.0 – 0.0]	71.3 [48.1 – 95.8]
13-17	0.2 [0.1 – 0.4]	4.1 [2.5 – 6.6]	15.3 [9.4 – 24.1]	25.3 [16.9 – 37.3]	10.3 [6.3 – 16.5]	5.5 [3.4 – 8.9]	0.0 [-0.0 – 0.0]	50.5 [33.9 – 74.1]
18-49	0.3 [0.2 – 0.4]	4.9 [3.2 – 6.8]	15.5 [10.2 – 21.6]	9.5 [6.3 – 13.3]	16.5 [11.5 – 22.5]	6.3 [4.2 – 8.9]	0.0 [-0.0 – 0.0]	45.3 [32.4 – 59.9]
50-64	0.6 [0.4 – 0.8]	10.4 [6.8 – 14.3]	34.4 [22.6 – 47.4]	21.8 [14.4 – 30.1]	26.8 [17.7 – 37.0]	33.5 [23.8 – 44.3]	0.0 [-0.0 – 0.0]	104.7 [74.6 – 138.6]
65+	4.8 [3.5 – 6.0]	80.9 [59.4 – 102.7]	261.5 [190.6 – 332.0]	163.1 [118.4 – 208.2]	195.0 [142.7 – 249.0]	115.7 [85.7 – 146.5]	203.5 [171.0 – 235.0]	776.3 [621.4 – 927.4]
All	1.5 [1.1 – 1.8]	23.0 [17.4 – 28.0]	67.3 [49.1 – 84.1]	41.3 [29.8 – 52.6]	50.6 [36.7 – 63.9]	31.8 [23.5 – 39.7]	35.5 [29.9 – 41.1]	194.3 [154.7 – 228.8]

Hospitalizations averted per 100,000 doses

Each cell divides hospitalizations averted in the **row's** age group by the doses scheduled for the **column's** age group, so off-diagonal cells are the indirect benefit per dose spent on the vaccinated group. The **All vaccinated** column has no single targeted group, so there each row keeps its own age group's dose count, matching Table S.A.1's per-dose columns.

Age group (counted)	0 vaccinated	1-4 vaccinated	5-12 vaccinated	13-17 vaccinated	18-49 vaccinated	50-64 vaccinated	65+ vaccinated	All vaccinated
0	63.0 [40.9 – 82.6]	4.0 [2.4 – 5.2]	6.7 [4.1 – 8.7]	7.3 [4.4 – 9.6]	1.8 [1.1 – 2.3]	1.2 [0.7 – 1.6]	0.0 [-0.0 – 0.0]	259.9 [167.7 – 339.5]
1-4	9.7 [6.9 – 11.5]	96.6 [71.9 – 116.7]	37.9 [27.0 – 44.9]	37.4 [26.4 – 44.3]	8.8 [6.3 – 10.5]	6.1 [4.3 – 7.3]	0.0 [-0.0 – 0.0]	188.8 [140.6 – 228.1]
5-12	6.7 [4.2 – 9.5]	16.4 [10.2 – 23.0]	63.7 [42.4 – 85.9]	33.8 [20.7 – 47.9]	7.0 [4.3 – 10.0]	5.0 [3.0 – 7.1]	0.0 [-0.0 – 0.0]	99.8 [67.4 – 134.1]
13-17	3.0 [1.9 – 4.9]	6.7 [4.1 – 10.7]	14.6 [8.9 – 22.9]	45.8 [30.5 – 67.4]	3.5 [2.1 – 5.6]	2.6 [1.6 – 4.2]	0.0 [-0.0 – 0.0]	91.3 [61.3 – 134.0]
18-49	27.6 [18.3 – 38.3]	57.2 [38.0 – 79.9]	106.7 [70.4 – 148.7]	124.4 [81.9 – 174.3]	40.8 [28.4 – 55.4]	22.0 [14.4 – 30.8]	0.0 [-0.0 – 0.0]	111.7 [80.0 – 147.7]
50-64	27.0 [17.7 – 37.5]	58.2 [38.2 – 80.4]	113.1 [74.4 – 155.8]	136.6 [89.7 – 188.1]	31.6 [20.9 – 43.7]	55.5 [39.4 – 73.4]	0.0 [-0.0 – 0.0]	173.5 [123.6 – 229.7]
65+	182.7 [134.5 – 231.7]	388.9 [285.2 – 493.4]	737.3 [537.5 – 936.2]	874.6 [634.8 – 1116.1]	197.3 [144.4 – 252.0]	164.3 [121.7 – 208.1]	278.0 [233.6 – 321.1]	1060.4 [848.8 – 1266.8]
All	320.1 [241.0 – 387.8]	631.9 [478.7 – 769.3]	1086.6 [793.3 – 1356.9]	1266.2 [915.8 – 1614.5]	293.2 [212.8 – 370.3]	258.3 [191.3 – 322.8]	278.0 [233.6 – 321.1]	347.6 [276.8 – 409.4]

Off-diagonal entries confirm real indirect effects — e.g. vaccinating 5-12 alone reduces hospitalizations in 0 by 24.4% and in 1-4 by 25.8%, both larger than several of those groups' own-age direct effects, consistent with school-age children acting as a major transmission hub in the contact structure. 65+ is the only group with zero indirect effect on every other group.

Per dose, the indirect benefit is dominated by what it does for **65+**: vaccinating 13-17 averts 874.6 [634.8 – 1116.1] hospitalizations per 100,000 doses in the 65+ group alone, against 45.8 [30.5 – 67.4] in 13-17 itself. This is why the **All** row — hospitalizations averted across the whole population per dose — ranks 13-17 (1266.2) and 5-12 (1086.6) far above the groups that carry the burden directly.

Table S.A.4 — Vaccine-effectiveness sensitivity scenarios

Implied vaccine effectiveness under three illustrative VE presets — a parameter table, not a simulation result. **Baseline VE (fitted)** is the model's fitted VE (included here as the reference point, not itself a sensitivity scenario); **Low VE/High VE** bracket it and correspond to the estimated lower and upper bound of VE. The ratio of low/high VE to baseline remains constant across parameter sets.

Scenario	Age group	VE against infection	VE against hospitalization (overall)	VE against hospitalization, given infection
Low VE	0	33%	33%	0%
Low VE	1-4	33%	33%	0%
Low VE	5-12	33%	33%	0%
Low VE	13-17	33%	33%	0%

Scenario	Age group	VE against infection	VE against hospitalization (overall)	VE against hospitalization, given infection
Low VE	18-49	6%	10%	4%
Low VE	50-64	6%	10%	4%
Low VE	65+	0%	21%	21%
Baseline VE (fitted)	0	43%	48%	9%
Baseline VE (fitted)	1-4	43%	48%	9%
Baseline VE (fitted)	5-12	43%	48%	9%
Baseline VE (fitted)	13-17	43%	48%	9%
Baseline VE (fitted)	18-49	21%	28%	9%
Baseline VE (fitted)	50-64	21%	28%	9%
Baseline VE (fitted)	65+	0%	31%	31%
High VE	0	51%	78%	55%
High VE	1-4	51%	78%	55%
High VE	5-12	51%	78%	55%
High VE	13-17	51%	78%	55%
High VE	18-49	34%	44%	16%
High VE	50-64	34%	44%	16%
High VE	65+	14%	39%	29%

Also note **65+ has 0% VE against infection in the fitted baseline and in Low VE** — residual susceptibility is 1.00 for that age group in both (§2.2), so only the severity-protection channel operates there. **High VE** is the exception: it scales residual susceptibility to 0.86, giving 65+ a 14% VE against infection, so that scenario is the only one in which vaccinating 65+ blocks any transmission at all. This is why 65+ shows an all-zero indirect effect on every other age group throughout Table S.A.2, which is built on the fitted baseline.

Table S.A.5 — Hospitalizations averted across VE scenarios

Compares each VE sensitivity scenario (§ above) against no vaccination at all.

% reduction in hospitalizations

Age group	Low VE	Baseline VE (fitted)	High VE
0	48.6% [43.5% – 52.3%]	69.6% [64.6% – 73.0%]	82.9% [79.6% – 85.2%]
1-4	55.6% [50.7% – 59.3%]	75.8% [71.5% – 78.8%]	89.1% [86.8% – 90.6%]
5-12	52.7% [46.8% – 57.0%]	72.8% [67.4% – 76.5%]	85.9% [82.7% – 88.1%]
13-17	49.2% [43.1% – 53.7%]	70.0% [64.1% – 74.0%]	83.6% [79.8% – 86.1%]
18-49	41.3% [35.3% – 45.8%]	64.8% [58.6% – 69.0%]	78.5% [73.8% – 81.5%]

Age group	Low VE	Baseline VE (fitted)	High VE
50-64	42.2% [36.5% – 46.4%]	66.6% [60.9% – 70.5%]	80.4% [76.3% – 83.0%]
65+	48.4% [43.6% – 52.0%]	69.7% [64.8% – 72.9%]	81.9% [78.4% – 84.2%]
All	47.3% [42.3% – 51.1%]	69.1% [64.1% – 72.5%]	81.8% [78.2% – 84.1%]

Hospitalizations averted per 100,000 population

Age group	Low VE	Baseline VE (fitted)	High VE
0	82.4 [52.1 – 106.9]	117.9 [76.0 – 154.0]	140.1 [90.7 – 183.4]
1-4	126.2 [92.1 – 150.1]	171.2 [127.5 – 206.9]	201.2 [150.3 – 244.1]
5-12	51.3 [33.4 – 69.8]	71.3 [48.1 – 95.8]	84.1 [58.5 – 112.2]
13-17	35.3 [23.1 – 53.2]	50.5 [33.9 – 74.1]	60.6 [41.4 – 87.1]
18-49	28.9 [19.7 – 39.3]	45.3 [32.4 – 59.9]	54.9 [40.9 – 71.5]
50-64	66.2 [45.1 – 89.7]	104.7 [74.6 – 138.6]	126.3 [91.8 – 164.7]
65+	539.2 [414.0 – 659.2]	776.3 [621.4 – 927.4]	912.4 [748.1 – 1073.2]
All	133.0 [102.4 – 161.5]	194.3 [154.7 – 228.8]	229.4 [188.5 – 266.7]

Even under the pessimistic **Low VE** assumption, the fitted vaccination schedule still averts ~47% of hospitalizations overall — the schedule's *coverage* (§2.5) matters roughly as much as the assumed per-dose effectiveness across this VE range.

4. Baseline fit check — posterior-uncertainty simulation vs. raw data

This section simulates the baseline (fitted-vaccination) scenario once per posterior parameter draw — all 638 draws (§2.3) — and reports the median and 95% interval across draws, to check the calibrated model against the raw data it was fit to.

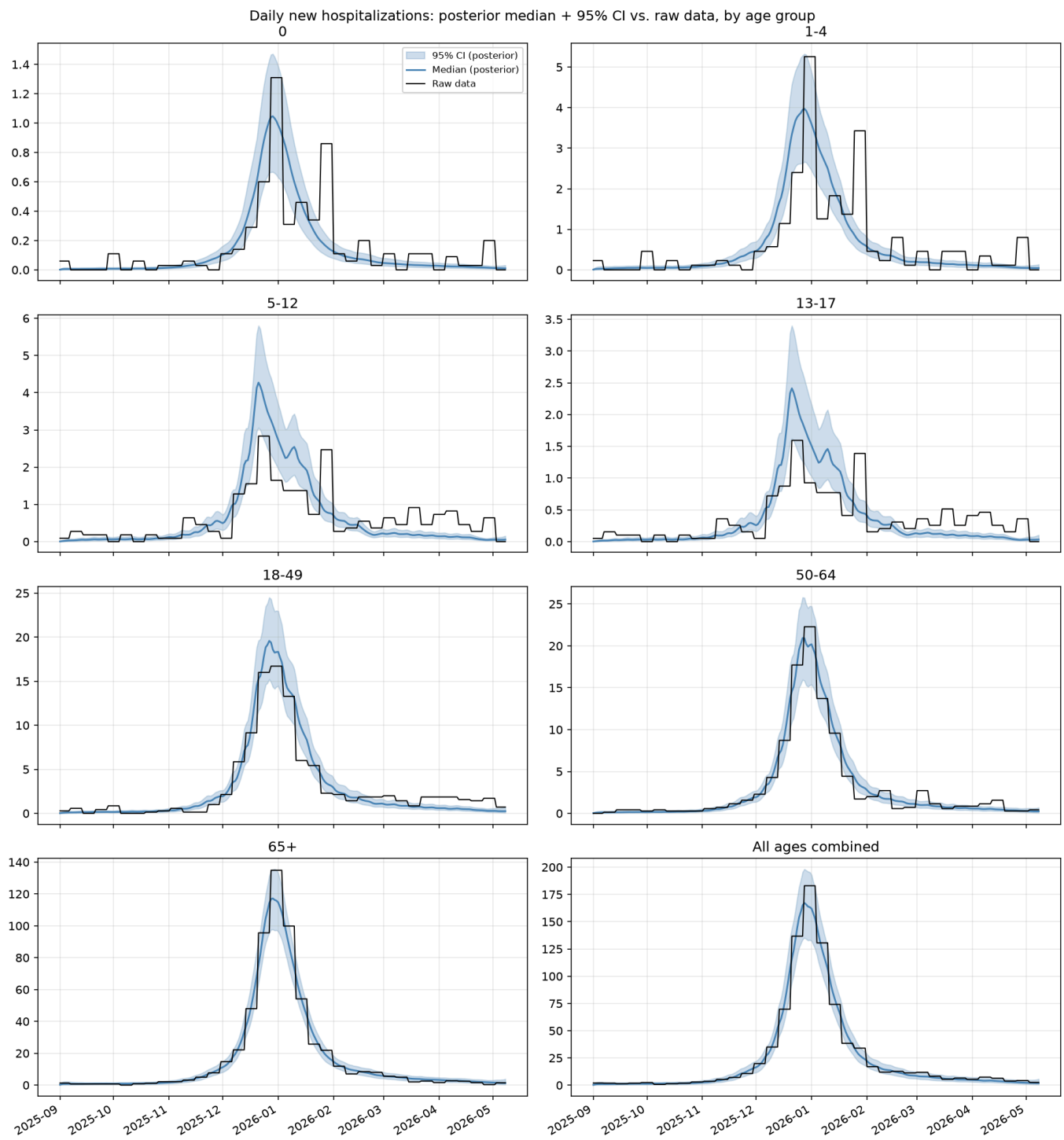
Cumulative hospitalizations, by age group

Sum of simulated (posterior median and 95% interval) vs. raw daily hospital admissions, over the range of dates common to both series (2025-09-08 – 2026-05-17):

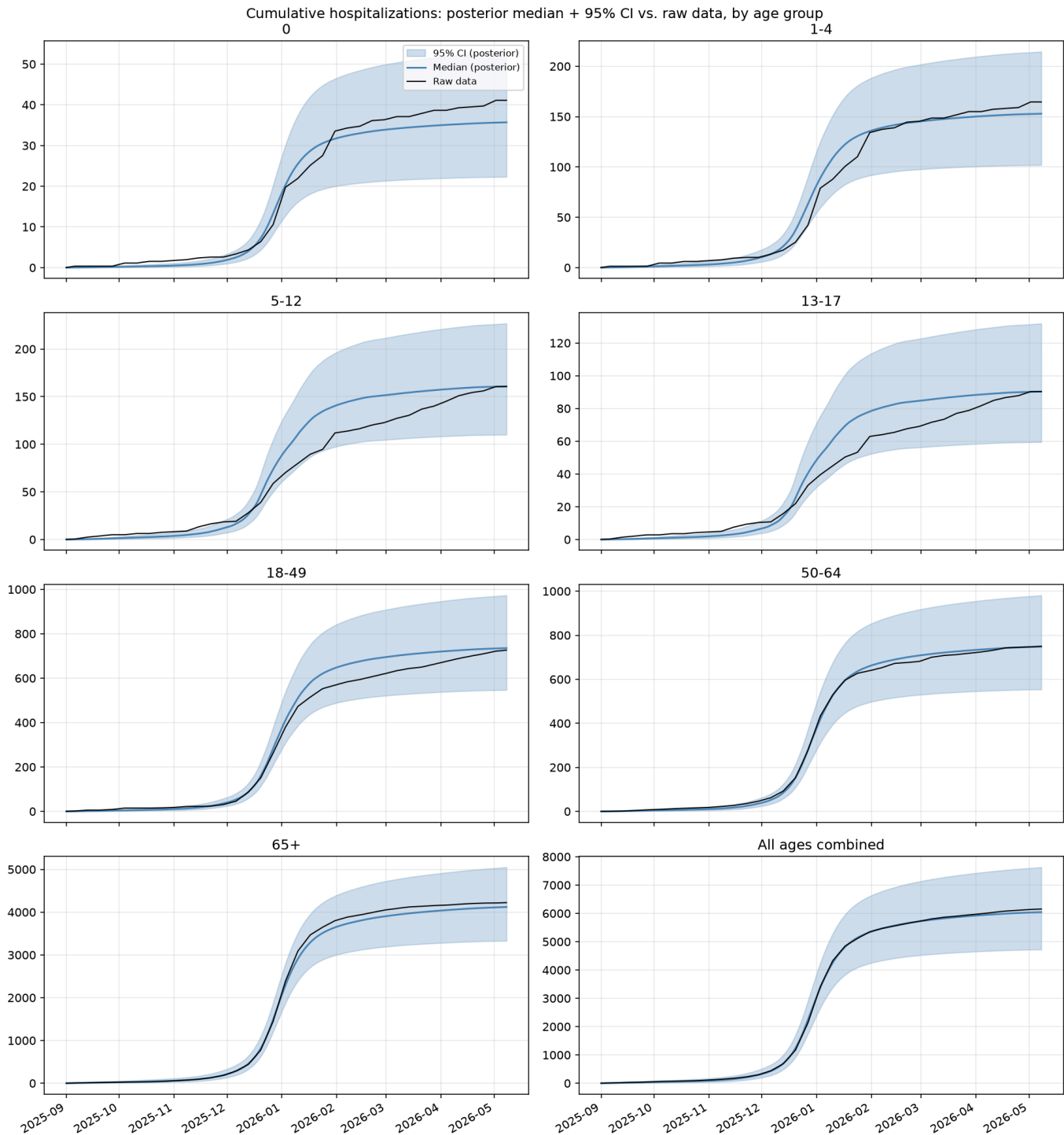
Age group	Simulated (median)	Simulated 95% interval	Raw data	% difference (median)
0	35.7	22.3 – 53.1	41.1	-13.1%
1-4	152.9	102.0 – 214.5	164.6	-7.1%
5-12	160.8	110.1 – 226.8	160.5	0.2%
13-17	90.3	59.5 – 131.9	90.3	0.0%
18-49	735.2	546.7 – 972.8	726.0	1.3%
50-64	748.5	553.8 – 981.0	749.6	-0.1%
65+	4119.7	3328.8 – 5049.5	4222.3	-2.4%
All	6043.1	4723.1 – 7629.5	6154.4	-1.8%

The fit tracks the data closely overall (-1.8% on the total, raw value within the 95% interval for every age group) — the largest relative miss is in the smallest group (age 0, -13.1%, but only ~5 admissions off in absolute terms).

Daily new hospitalizations by age group



Cumulative hospitalizations by age group



The posterior interval captures the single main epidemic wave (peaking around late December/early January) well across all age groups.

5. Data sources

Input	Source
Age-specific vaccination coverage	MIDAS Flu Scenario Modeling Hub resources, age-specific coverage dataset
Hospital admissions (calibration target)	MIDAS Flu Scenario Modeling Hub, target-data time series
Population by age group	US Census (tidycensus)
Contact matrices	Mistry et al. 2021 synthetic contact matrices
Absolute humidity	gridMET daily specific-humidity data, averaged over Massachusetts

Input	Source
School/work calendar	Constructed school/work-day calendar for the state, for each day of the season

6. Notes

- **The confidence intervals in §3 reflect calibration uncertainty, not epidemic-process noise** — they come from re-running each scenario under 638 different (but similarly plausible) posterior parameter draws, not from chain-binomial simulation noise at fixed parameters.
- **$m(t)$ is a statistical smoothing device, not a mechanistic term** — the 14-day-knot random walk in this fit absorbs whatever transmission variation the mechanistic model (contacts, humidity) doesn't explain; it should not be read as an independently-measured behavioral signal.
- **The "scale to 70% coverage" scenarios (Table S.A.3/S.A.6, appendix) treat 70% as a floor, not a quota** — age groups already above it (1-4 at 90.7%, 5-12 at 71.4%, 65+ at 73.2%) keep their baseline schedule untouched rather than having uptake cut back, so they contribute all-zero columns.
- **"Per 100,000 doses" counts doses the schedule reports, not doses the model delivers** — 3.3% of scheduled doses go to people already infected (see the dose-accounting appendix). Those are wasted, not unspent, so they belong in a cost-effectiveness denominator; excluding them would flatter each scenario in proportion to how large its epidemic was.

Appendix: Tables S.A.3 and S.A.6

Table S.A.3 — Additional hospitalizations averted at 70% coverage

Each column scales a single age group's vaccination schedule up to 70% cumulative coverage; "All" scales every eligible age group. Compared against the baseline vaccination scenario. Per §2.5, 70% is treated as a **floor**: the four groups below it (0, 13-17, 18-49, 50-64) are scaled up, and the three already above it (1-4, 5-12, 65+) keep their baseline schedule rather than being de-vaccinated. Those three are therefore identical to baseline and give exact-zero columns.

% reduction in hospitalizations

Age group (counted)	0 vaccinated	1-4 vaccinated	5-12 vaccinated	13-17 vaccinated	18-49 vaccinated	50-64 vaccinated	65+ vaccinated	All vaccinated
0	11.2% [11.0% – 11.4%]	0.0% [0.0% – 0.0%]	0.0% [0.0% – 0.0%]	5.7% [5.2% – 6.1%]	19.3% [17.8% – 20.3%]	2.3% [2.1% – 2.4%]	0.0% [0.0% – 0.0%]	34.0% [32.3% – 35.3%]
1-4	0.4% [0.4% – 0.4%]	0.0% [0.0% – 0.0%]	0.0% [0.0% – 0.0%]	5.6% [5.1% – 6.1%]	18.3% [16.9% – 19.4%]	2.2% [2.0% – 2.4%]	0.0% [0.0% – 0.0%]	25.1% [23.3% – 26.4%]
5-12	0.3% [0.3% – 0.4%]	0.0% [0.0% – 0.0%]	0.0% [0.0% – 0.0%]	5.9% [5.2% – 6.3%]	17.0% [15.4% – 18.2%]	2.1% [1.9% – 2.3%]	0.0% [0.0% – 0.0%]	23.9% [21.9% – 25.4%]
13-17	0.3% [0.3% – 0.3%]	0.0% [0.0% – 0.0%]	0.0% [0.0% – 0.0%]	14.2% [13.4% – 14.8%]	17.2% [15.5% – 18.4%]	2.2% [1.9% – 2.4%]	0.0% [0.0% – 0.0%]	30.9% [28.8% – 32.4%]
18-49	0.4% [0.3% – 0.4%]	0.0% [0.0% – 0.0%]	0.0% [0.0% – 0.0%]	5.8% [5.1% – 6.2%]	24.1% [22.5% – 25.2%]	2.4% [2.2% – 2.6%]	0.0% [0.0% – 0.0%]	30.6% [28.6% – 32.0%]
50-64	0.3% [0.3% – 0.3%]	0.0% [0.0% – 0.0%]	0.0% [0.0% – 0.0%]	5.8% [5.2% – 6.2%]	18.7% [17.2% – 19.8%]	4.8% [4.6% – 5.0%]	0.0% [0.0% – 0.0%]	27.6% [25.6% – 28.9%]
65+	0.3% [0.3% – 0.4%]	0.0% [0.0% – 0.0%]	0.0% [0.0% – 0.0%]	5.8% [5.3% – 6.2%]	18.5% [17.1% – 19.8%]	2.6% [2.3% – 2.7%]	0.0% [0.0% – 0.0%]	25.6% [23.9% – 27.3%]

Age group (counted)	0 vaccinated	1-4 vaccinated	5-12 vaccinated	13-17 vaccinated	18-49 vaccinated	50-64 vaccinated	65+ vaccinated	All vaccinated
					19.5%]			26.9%]
All	0.4% [0.4% – 0.4%]	0.0% [0.0% – 0.0%]	0.0% [0.0% – 0.0%]	5.9% [5.3% – 6.4%]	19.2% [17.6% – 20.2%]	2.8% [2.6% – 3.0%]	0.0% [0.0% – 0.0%]	26.6% [24.7% – 27.9%]

Hospitalizations averted per 100,000 population

Age group (counted)	0 vaccinated	1-4 vaccinated	5-12 vaccinated	13-17 vaccinated	18-49 vaccinated	50-64 vaccinated	65+ vaccinated	All vaccinated
0	5.8 [3.7 – 8.0]	0.0 [0.0 – 0.0]	0.0 [0.0 – 0.0]	2.9 [1.9 – 4.0]	9.9 [6.5 – 13.5]	1.2 [0.8 – 1.6]	0.0 [0.0 – 0.0]	17.5 [11.4 – 24.1]
1-4	0.2 [0.2 – 0.3]	0.0 [0.0 – 0.0]	0.0 [0.0 – 0.0]	3.1 [2.2 – 3.9]	10.0 [7.2 – 12.8]	1.2 [0.9 – 1.5]	0.0 [0.0 – 0.0]	13.7 [9.8 – 17.6]
5-12	0.1 [0.1 – 0.1]	0.0 [0.0 – 0.0]	0.0 [0.0 – 0.0]	1.6 [1.1 – 2.0]	4.5 [3.3 – 5.9]	0.6 [0.4 – 0.7]	0.0 [0.0 – 0.0]	6.4 [4.6 – 8.3]
13-17	0.1 [0.0 – 0.1]	0.0 [0.0 – 0.0]	0.0 [0.0 – 0.0]	3.1 [2.2 – 4.2]	3.8 [2.6 – 5.1]	0.5 [0.3 – 0.7]	0.0 [0.0 – 0.0]	6.8 [4.7 – 9.2]
18-49	0.1 [0.1 – 0.1]	0.0 [0.0 – 0.0]	0.0 [0.0 – 0.0]	1.4 [1.1 – 1.8]	5.9 [4.8 – 7.3]	0.6 [0.5 – 0.7]	0.0 [0.0 – 0.0]	7.6 [6.1 – 9.3]
50-64	0.2 [0.1 – 0.2]	0.0 [0.0 – 0.0]	0.0 [0.0 – 0.0]	3.1 [2.4 – 3.8]	9.9 [7.7 – 12.3]	2.6 [2.0 – 3.2]	0.0 [0.0 – 0.0]	14.6 [11.3 – 18.0]
65+	1.1 [1.0 – 1.3]	0.0 [0.0 – 0.0]	0.0 [0.0 – 0.0]	19.6 [17.3 – 21.9]	62.4 [56.1 – 69.2]	8.7 [7.7 – 9.6]	0.0 [0.0 – 0.0]	86.7 [78.0 – 96.0]
All	0.4 [0.3 – 0.4]	0.0 [0.0 – 0.0]	0.0 [0.0 – 0.0]	5.1 [4.6 – 5.6]	16.6 [15.1 – 18.1]	2.4 [2.2 – 2.7]	0.0 [0.0 – 0.0]	23.1 [21.1 – 25.1]

Hospitalizations averted per 100,000 additional doses

Denominators are the *additional* doses each scenario schedules, which reduces to $\max(0, 70\% - \text{baseline coverage}) \times \text{population}$: 17,270 doses for age 0, 60,431 for 13-17, 877,865 for 18-49, 137,410 for 50-64, and zero for the three groups already above 70% (shown as —). Being a property of the schedule, this is exact and identical across all 638 parameter draws. The **All ages** column raises four groups at once, so its denominator is the **total** 1,092,976 additional doses.

Because every column divides all of its rows by that one dose count, the age rows within a column decompose its **All** row by where the averted burden lands — and so sum to it, up to small differences from these cells being medians of per-replicate ratios rather than ratios of medians.

Age group (counted)	0 vaccinated	1-4 vaccinated	5-12 vaccinated	13-17 vaccinated	18-49 vaccinated	50-64 vaccinated	65+ vaccinated	All vaccinated
0	23.4 [15.2 – 32.4]	—	—	3.4 [2.3 – 4.6]	0.8 [0.5 – 1.1]	0.6 [0.4 – 0.8]	—	1.1 [0.7 – 1.5]
1-4	3.4 [2.5 – 4.4]	—	—	14.3 [10.4 – 18.0]	3.2 [2.3 – 4.1]	2.5 [1.8 – 3.1]	—	3.5 [2.5 – 4.5]
5-12	3.1 [2.2 – 4.0]	—	—	15.6 [11.4 – 20.5]	3.1 [2.3 – 4.1]	2.5 [1.8 – 3.2]	—	3.5 [2.6 – 4.6]

Age group (counted)	0 vaccinated	1-4 vaccinated	5-12 vaccinated	13-17 vaccinated	18-49 vaccinated	50-64 vaccinated	65+ vaccinated	All vaccinated
13-17	1.6 [1.1 – 2.2]	—	—	21.3 [14.9 – 28.8]	1.8 [1.2 – 2.4]	1.4 [1.0 – 2.0]	—	2.6 [1.8 – 3.5]
18-49	15.1 [11.9 – 18.8]	—	—	70.3 [55.0 – 88.0]	20.2 [16.2 – 24.9]	12.9 [10.2 – 16.1]	—	20.6 [16.6 – 25.4]
50-64	14.2 [10.9 – 17.7]	—	—	72.0 [55.5 – 90.1]	16.1 [12.5 – 20.0]	26.6 [20.8 – 32.7]	—	19.0 [14.7 – 23.5]
65+	81.0 [72.0 – 89.9]	—	—	397.1 [349.2 – 442.7]	86.9 [78.0 – 96.3]	77.0 [68.6 – 85.5]	—	96.9 [87.1 – 107.3]
All	142.4 [129.2 – 155.0]	—	—	595.5 [530.4 – 652.0]	132.4 [120.1 – 144.3]	124.0 [111.5 – 135.2]	—	147.5 [134.7 – 160.4]

Per additional dose, raising **13-17** to 70% is far and away the best buy: 595.5 [530.4 – 652.0] hospitalizations averted per 100,000 doses, against 142.4 for age 0 and 132.4 for 18-49 — and 397.1 of that 595.5 lands in **65+**, not in 13-17 itself. 18-49 dominates the *absolute* totals only because it absorbs 877,865 of the 1,092,976 additional doses.

Table S.A.6 — Additional hospitalizations averted at 70% coverage, across VE scenarios

For each VE sensitivity scenario, compares that scenario's own baseline vaccination to the 70%-coverage floor applied to every eligible age group.

% reduction in hospitalizations

Age group	Low VE	Baseline VE (fitted)	High VE
0	16.3% [14.8% – 17.4%]	34.0% [32.3% – 35.3%]	48.4% [46.9% – 49.6%]
1-4	9.7% [8.2% – 10.8%]	25.1% [23.3% – 26.4%]	33.4% [31.7% – 34.9%]
5-12	9.3% [7.7% – 10.5%]	23.9% [21.9% – 25.4%]	31.8% [30.0% – 33.5%]
13-17	14.5% [12.7% – 15.9%]	30.9% [28.8% – 32.4%]	42.7% [40.9% – 44.3%]
18-49	11.7% [10.0% – 12.9%]	30.6% [28.6% – 32.0%]	42.8% [41.3% – 44.1%]
50-64	10.5% [8.9% – 11.6%]	27.6% [25.6% – 28.9%]	38.6% [37.1% – 40.0%]
65+	9.9% [8.4% – 10.9%]	25.6% [23.9% – 26.9%]	35.9% [34.4% – 37.1%]
All	10.3% [8.7% – 11.3%]	26.6% [24.7% – 27.9%]	37.1% [35.6% – 38.4%]

Hospitalizations averted per 100,000 population

Age group	Low VE	Baseline VE (fitted)	High VE
0	14.2 [9.2 – 18.7]	17.5 [11.4 – 24.1]	14.0 [9.1 – 19.9]
1-4	9.7 [7.2 – 11.8]	13.7 [9.8 – 17.6]	8.3 [5.7 – 10.9]
5-12	4.3 [2.9 – 5.8]	6.4 [4.6 – 8.3]	4.4 [3.2 – 5.8]
13-17	5.4 [3.7 – 7.5]	6.8 [4.7 – 9.2]	5.1 [3.6 – 7.0]
18-49	4.8 [3.6 – 6.2]	7.6 [6.1 – 9.3]	6.5 [5.2 – 7.9]
50-64	9.4 [6.8 – 12.4]	14.6 [11.3 – 18.0]	12.0 [9.6 – 14.6]
65+	56.7 [45.5 – 66.6]	86.7 [78.0 – 96.0]	72.4 [65.4 – 79.3]

Age group	Low VE	Baseline VE (fitted)	High VE
All	15.2 [12.2 – 17.5]	23.1 [21.1 – 25.1]	19.0 [17.6 – 20.6]

Appendix: dose accounting — scheduled vs. delivered doses

Two different dose counts appear in this model, and the per-100,000-doses panels depend on which one is used.

Scheduled doses are what the vaccination schedule reports: the daily vaccination proportions of §2.5, summed over the season and multiplied by population. **Delivered** doses are what the model records as an actual $S \rightarrow SV$ transition. They differ because of the cap in §1.4 — a dose is only delivered if the intended recipient is still in S . Someone already infected, recovered, or hospitalized is skipped.

That gap is not a saving. In the real world the dose is still bought, shipped and injected; it simply arrives after the recipient has already been infected and buys no protection. Counting only delivered doses would therefore understate the true cost of every scenario — and understate it *most* for the scenarios with the largest epidemics, which is exactly backwards for a cost-effectiveness denominator. Every "per 100,000 doses" figure in this report divides by the scheduled count.

Baseline schedule, median across the 638 posterior draws:

Age group	Population	Scheduled doses	Delivered doses	Wasted doses	% wasted	Scheduled coverage	Delivered coverage
0	70,067	31,777	31,409	368	1.2%	45.4%	44.8%
1-4	280,268	254,216	251,202	3,015	1.2%	90.7%	89.6%
5-12	606,291	433,128	421,409	11,719	2.7%	71.4%	69.5%
13-17	411,782	227,816	222,816	5,000	2.2%	55.3%	54.1%
18-49	2,978,204	1,206,878	1,154,627	52,251	4.3%	40.5%	38.8%
50-64	1,424,434	859,693	827,816	31,877	3.7%	60.4%	58.1%
65+	1,221,349	894,101	868,876	25,225	2.8%	73.2%	71.1%
All	6,992,395	3,907,610	3,778,154	129,456	3.3%	55.9%	54.0%

Waste tracks infection attack rate, as expected: it is lowest in the youngest groups (1.2% in 0 and 1-4) and highest in 18-49 (4.3%) and 50-64 (3.7%), which between them account for 84,128 of the 129,456 wasted doses. Overall 3.3% of the season's scheduled doses land in arms that no longer benefit, pulling realized coverage from 55.9% down to 54.0%.

A second consequence, relevant to why the per-dose tables are built the way they are: delivered doses are an *output* of the simulation, so they vary with the parameter draw even when the schedule is identical — by 1.2% to 2.9% across the posterior — and they leak between age groups, since a milder epidemic in one group leaves more susceptibles for the cap to reach in another. Scheduled doses have neither property.